Cheatsheet: Creating an Exercise Sheet

From installation to multiple output formats

OUTPUT	STRUCTURE	BEST FOR
One file (HTML / PDF / Word)	Single `.qmd`, no project needed	Handouts, homework, printables

1 — Setup

Install (once per computer):

- **VS Code** — code.visualstudio.com (tick **"Add to PATH"** during setup)
- **Quarto CLI** — quarto.org/docs/get-started
- **Quarto extension** for VS Code — search "Quarto" in the Extensions sidebar

> ℹ Only if you want PDF output
>
> `quarto install tinytex` in the Terminal — installs the LaTeX engine Quarto needs for PDF. Skip this if you only want HTML or Word.

Create the file: File → New File... → Quarto Document → save it as e.g. `exercises.qmd`. Unlike a website, this is a single file — no project folder structure needed.

Which one do I need?

Need one self-contained document to print, email, or hand out — with a title, deadline, and export to PDF or Word? This is it. For a full multi-page site, see the **Website** cheatsheet; for a long document with numbered chapters, see the **Book** cheatsheet.

2 — Writing your exercise sheet

Page header sets the title, author, and output format:

```
---
title: "Exercise Sheet 1"
author: "Your Name"
format: pdf
---
```

Body: use `##` for each exercise, plain paragraphs for the text, `$...$` for inline math:

```
## Exercise 1

Calculate the derivative of $f(x) = x^2 + 3x$.
```

Code examples: a plain fence just *displays* code without running it — good for "here's what the answer looks like":

```
x = [1, 2, 3, 4]
print(sum(x))
```

> **Want it to actually run?**
>
> Tag the fence `{python}` instead of `python` — it executes via Jupyter and the output is embedded right in the PDF or Word file, no browser needed. That's different from the `pyodide-interaktiv` cells, which only run live in a browser (see the Extensions cheatsheet).

Preview vs. Render (same idea as for a website, just for one file):

Command	What it does
`quarto preview exercises.qmd`	Live browser view, updates on save
`quarto render exercises.qmd`	Builds the final output file once

3 — Output formats

One `.qmd` file, several possible outputs — change the `format:` line:

Format	YAML value	Needs
HTML	`format: html`	Nothing extra
Word	`format: docx`	Nothing extra
PDF	`format: pdf`	TinyTeX (page 1)

All formats at once:

```yaml
---
title: "Exercise Sheet 1"
format:
  html: default
  pdf: default
  docx: default
---
```

Run `quarto render` once — Quarto produces all three files in the same folder.

Good to know

- Only need one of several configured formats right now? `quarto render exercises.qmd --to pdf` renders just that one.
- `{{< pagebreak >}}` on its own line forces a new page — handy to keep one exercise per printed page.
- Want your own letterhead or styles in the `.docx` output? `format: docx: reference-doc: template.docx` applies a Word template.

Full details & screenshots: What you need · Create an Exercise Sheet · Other Output Formats